TOMMASO ISIDORI

CERN
Espl. des Particules 1 ⋄ 1211 Meyrin (CH)
+33 766749890 ⋄ tommaso.isidori@cern.ch ⋄ LinkedIn profile

SUMMARY

I am an experimental physicist with a strong background in particle detector design, assembly, and testing, particularly in the context of high-energy and nuclear physics experiments. As a postdoctoral researcher at the University of Kansas, I am based at CERN in Geneva, where I coordinate test campaigns and prototype performance for the ALICE FoCal upgrade project. I am deeply involved in hardware optimization and customized readout electronics. I develop DAQ systems using Python, C++, and the pyVisa library for instrumentation control. I also design and implement data acquisition frameworks, and integrate hardware components within the ALICE O2 readout and reconstruction infrastructure using ROOT and Streamlit for data quality monitoring and user interfaces. Beyond FoCal, I am further involved in the ALICE collaboration for the operation of the EMCAL detector, holding the position of Technical Coordinator and System Run Coordinator. I also contribute to the ePIC Collaboration as part of the Electron-Ion Collider (EIC) efforts, where I conduct R&D on fast, radiation-hard timing detectors, including fast diamonds, and silicon sensors, for electron Compton scattering polarimetry. I am proficient in the use C/C++, CERN software (ROOT, GEANT4, O2,..) and Python for data analysis (NumPy, pandas, polars, pyCharm, VS-code, ...) and I collaborate with computer science students to develop algorithms for signal processing.

CURRENT POSITION

Postdoctoral researcher at the University of Kansas *2021 - present*

Member of the ALICE collaboration

Member of the ePIC collaboration

Prototype performance coordinator of the ALICE FoCal project

Technical Coordinator and System Run Coordinator of the ALICE EMCAL project

Lab responsible for the ALICE FoCal project

Lab responsible for the ALICE EMCAL project

Based at CERN

EDUCATION

Ph.D. in Physics *2017 - 2022*

The University of Kansas (KU), Lawrence, Kansas, U.S.

Thesis: *New Generation Fast Silicon Detectors for Timing and Particle Identification: High Energy Physics and Applications*

Supervisor: Christophe Royon

Co-Supervisor: Nicola Minafra

Master's Degree in Physics, Foundamental Interaction *2014 - 2017*

Universitá degli studi di Pisa, Pisa, Italy.

Thesis: *Diamond detectors for proton Time of Flight in CT-PPS and TOTEM experiments*

Supervisor: Nicola Turini, Simone Donati

Bachelor's Degree in Physics *2010 - 2014*

Universitá degli studi di Siena, Siena, Italy.

Thesis: *Characterization and Optimization of Gas Detectors*

Supervisor: Nicola Turini

AWARDS

Recipient of the 2019 Graduate Travel Grant from the Department of Physics and Astronomy at the University of Kansas.

Awarded 2019 Best Ph.D. Talk at the 150th Kansas Academy of Sciences Meeting.

Awarded the 2017 KU Graduate Fellowship for the 2017/2018 academic year based on exceptional qualifications.

RESEARCH EXPERIENCE

ALICE Experiment 2022 - Present

FoCal Test beam and prototypes performance coordinator *CERN, CH*

- Coordinated teams, chaired weekly meetings, and co-organized data collection and analysis efforts. Scheduled events through CERN Indico to facilitate team meetings and follow up on results and organizational topics. Utilized Jira tickets to manage operations, track issues, and implement new features.
- Organized, coordinated, and participated in over ten test beam campaigns starting from Summer 2022. Responsibilities included managing material procurement and transportation via the CERN EDH system, maintaining communication with the PS/SPS CERN operation team both during and outside scheduled meetings, on-site assembly and operation of the setup and beam line, and instructing the team on beam operation and safety procedures.
- Coordinated laboratory activities including prototype assembly, testing, and characterization. Managed the installation, maintenance, and upgrades of the ALICE FoCal front-end chain components, which include the First Layer Processor (FLP), Common Readout Unit (CRU) modules, and Local Trigger Units (LTUs).
- Developed software tools (in Python and C++) for analyzing data from lab tests and calibrations. Created tools for instrumentation monitoring and control using microcontrollers (ESP32) for remote operation and monitoring of 18 layers of silicon pad detectors and 2 strings of high granularity ALPIDE MAPS used in the FoCal-E prototype tower.
- Developed a DAQ system (using C++, Python, Bash, XML) for the FoCal-E prototypes and integrated it within the ALICE Experiment Control System (ECS) framework. Contributed to the development of the FoCal Online-Offline (O2) infrastructure for real-time operation and monitoring of detector configurations and data streams.
- Acted as the interface between the FoCal project and the ALICE operations team for O2 development and hardware maintenance. Provided remote support and debugging for the FoCal teams during prototype testing.
- Contributed as a primary author to the approved FoCal Technical Design Report (TDR), specifically responsible for the Test Beams results chapter. Served as the main editor for three additional publications on the performance results of the FoCal prototype.

EMCAL Technical Coordination and System Run Coordination

- Since winter 2024: Trained in the coordination of the EMCAL operation team for data collection during proton and heavy ion runs. This include the expert level operation of the Detector Control System, Data Acquisition software, Quality Control, and Detector Safety Systems.
- Daily interface with the ALICE Run Coordination teams to coordinate and organize the detector operation and data collection, as well as the detector's hardware intervention during no-beam periods. Prompt reaction to the requests of the ALICE central teams during On-call shifts and offering expert support to the rest of the EMCAL operation team.
- Maintenance of the EMCAL DCS infrastructure through the management of the WinCC framework. Handling and maintenance of the detector trigger, DAQ, and Online quality monitor configurations stored in database.
- In charge of the On-call shifters training and mentorship for the independent operation of the EMCAL detector. Point of reference for the shifters for any major intervention needed during the ALICE data taking.

Detector operation

- Trained and operated the ALICE Detector Control System (DCS) and Detector Safety System (DSS) during dedicated shifts in the ALICE Control Room (CR). Provided support to the ALICE operation team by monitoring and managing the detector hardware using the specialized GUI developed by the collaboration. Acted as the liaison between the shift crew and the System Run Coordinators (SRC).

- Trained and operated the ALICE Experiment Control System (ECS) during dedicated shifts in the ALICE CR. This involved proficiency in loading DAQ partitions and detector configurations, monitoring the InfoLogger via the ECS GUI to address acquisition errors, logging details of ongoing runs, and tracking experiment performance using Grafana.
- Served as Shift Leader (SL) for the ALICE experiment. As a SL, bridged the communication between the Run Manager on duty and the shift crew in the ALICE CR, adhering to daily shift instructions to guide the experiment's operational routines.
- Trained and operated as Run Manager (RM) for the ALICE experiment. As a RM, collaborated with the ALICE Run Coordination (RC) to coordinate the central shifters, chaired and reported during daily meetings with the SRC and RC teams. Responsibilities included providing operational instructions to the three daily shift crews, logging and reporting issues, and maintaining close communication with ALICE and LHC teams. At the end of the RM period, prepared and presented a comprehensive report on operational efficiency and encountered issues to the RC and SRC teams.

Electron Polarimetry at the EIC 2021 - Present
R&D on timing detectors *CERN, CH*

- Conducted a market survey of advanced solid-state detector technologies including scCVD, pCVD, 3D diamond, LGAD, AC-LGAD, and 3DSi trenches. Managed the procurement of test equipment through CERN partner stores and external suppliers, including high-voltage (HV) and low-voltage (LV) power supplies, samplers, and monitoring tools.
- Designed, assembled, and tested radiation-hard fast timing detectors for high-rate particle beams. Responsibilities included selecting sensors, performing IV curve measurements, cleaning and wire bonding to custom fast electronics, conducting electrical tests and optimization of the readout, and evaluating performance with a Sr90 β emitting source.
- Utilized high-bandwidth and high-sampling-rate oscilloscopes, and developed a custom Python framework based on pyVisa for data acquisition (DAQ) and instrumentation control. This framework enables the selection of oscilloscope acquisition modes and parameters, remote initiation of acquisitions, and storage of waveforms in various formats (binary, data files, .hdf, .h5).
- Developed a Python-based signal analysis framework to convert collected waveforms into data frames (using pandas, polars, or serialized objects with pickle) for improved data handling. Implemented time-series data smoothing with custom algorithms and Python libraries (NumPy, SciPy), background rejection using statistical methods, and designed numerical and machine learning algorithms for pattern extraction and timing performance evaluation through Constant Fraction Discrimination (CFD) techniques.
- Mentored two computer science Ph.D. students at the University of Kansas in developing a comprehensive and organized repository for automating analysis steps and optimizing computational resources.

NASA AGILE project 2019 - 2022
Ph.D. project *Lawrence, KS*

- Selected and procured silicon sensors (manufactured by Micron) for the AGILE tracker. Utilized Weightfield2 software to simulate sensor responses to impinging particles. Measured IV characteristics and dark current using a semi-automatic probe station with tungsten probes at KU laboratories. Responsibilities included assembling sensors on custom electronics and characterizing them with an Am241 α radiation source.
- Procured and installed instrumentation such as power supplies, computers, and optical components to assemble a test bench for efficiency measurement using a fast pulsed laser.
- Developed a Python-based data acquisition (DAQ) chain using pyVisa and NI-VISA. Created a data acquisition framework in Jupyter notebooks for monitoring detector performance during laboratory tests.
- Tested the stability of power distribution boards based on CAEN A7508N modules. Conducted thermal and humidity cycle tests using a climate chamber installed at KU laboratories and utilized CAEN software for data acquisition.
- Finalized the development of a compact silicon tracker for ion species discrimination in space. This included a setup of three silicon detector layers connected to dual-gain readout electronics, featuring an on-board PSEC4 sampler chip and a Teensy 4.0 microcontroller.

CMS experiment Mip Timing Detector (MTD) 2019 - 2022
Ph.D. project *Lawrence, KS*

- Established a test center at KU for evaluating the performance of CMS LGAD wafer productions. I was involved in procuring and installing the infrastructure necessary for sensor testing, including setting up a system based on a semi-automatic probe station. This work encompassed the installation and maintenance of the chiller system and power distribution.
- Characterized LGAD production batches for the CMS Endcap Timing Layer (MTD-ETL). This involved controlling the movement of the testing chuck via a custom Python interface utilizing pattern recognition to identify sensor positions and establish electrical contact with the testing probes.
- Led KU USCMS efforts in over four test beam campaigns at the FERMILAB M-Test facilities. I installed a system based on a fast oscilloscope, controlled via a Python interface, to measure the timing performance of ETL LGADs. Collaborated with the KU analysis team to develop a ROOT framework, providing efficiency and timing performance feedback plots to the CMS MTD collaboration. The results are included in the MTD-ETL Technical Design Report (TDR).

Medical Physics 2017 - 2022
Ph.D. project *Lawrence, KS / Dublin, Ireland*

- Installed and wire-bonded an LGAD detector onto a custom readout board optimized for fast timing measurements. Characterized the detector using an Sr-90 β source and a fast oscilloscope.
- Installed the DUT and acquisition setup, including a fast oscilloscope and power distribution system, at St. Luke's Hospital, Dublin. Operated a medical linear accelerator (linac) as a high-rate electron source to evaluate the single-particle resolution capabilities of the LGAD.
- Designed a CERN ROOT analysis framework to decode oscilloscope files and organize data into n-tuples stored in ROOT TTree structures. The analysis was performed using a combination of ROOT, pyROOT, and Python-based scripts to implement data smoothing, background rejection, and peak detection.
- Lead author of the first publication on dosimetry and high-rate medical beam characterization using single-particle resolution measurements, published in *Physics in Medicine and Biology.*

Timing detector for the CT-PPS experiment 2016 - 2017
Master's degree project *CERN*

- Collaborated on the development of solid-state timing detectors (diamond, silicon) for the CMS-TOTEM forward tracker. My responsibilities included sensor selection and characterization, assembly on the readout board, and performing electrical tests. Additionally, I contributed to establishing the CT-PPS laboratory infrastructure at CERN and developing a stable testing framework.
- Participated in and led over 5 test beams at the H8 area at CERN NA, where the CT-PPS is installed. I worked on C++ software for data collection and reported results using CERN ROOT during daily and weekly TOTEM and CT-PPS meetings. I also became proficient in using the CERN CESAR beam operation software and organizing the beam schedule and requests.
- Led irradiation campaigns at the CERN East area IRRAD facilities to evaluate the performance degradation of diamond detectors subjected to integrated doses exceeding 10^{15} n_{eq}. I installed a bench setup using a GPIB-connected Keithley picoammeter and sourcemeter, controlled via C++ and C scripts, to monitor current absorption and efficiency loss of the DUT.
- Characterized the response of NINO fast discriminator chips by scanning the DAC threshold using the I2C protocol and sampling data through CAEN HPTDC modules to measure channel efficiency and timing performance.
- Led the characterization tests of a double-sided diamond prototype developed by the CT-PPS collaboration. To evaluate the efficacy of potting protection on detector boards, I tested prototypes using an X-ray gun setup to observe generated currents. These detectors were later installed at CERN NA for on-beam testing.
- Participated in the final installation of the current PPS timing diamond detectors inside the PPS Roman Pots (RP) in sectors 5-6 and 4-5 at the LHC.

Gas detectors for the TOTEM experiment Summer 2013
Bachelor's degree project *CERN*

- Summer student at the TOTEM experiment, collaborating with the CERN Gas Detector Development Group. Presented weekly updates on my progress.

- Conducted characterization and laboratory tests of Gas Electron Multipliers (GEM) detectors for the T2 inelastic tracker. Gained experience with NIM and VME modules, as well as standard lab instrumentation.
- Developed feedback systems for monitoring small current fluctuations (pA) in GEM layers. Collaborated with the electronics team to deepen my understanding of small current detection and processing using analog components.

ADDITIONAL EXPERIENCE

Referee Work

- Invited to review applications of the DOE Office of Science Graduate Student Research (SCGSR) Program's 2024 Solicitation.
- Invited for reviewing proceedings of the 16th Pisa Meeting on Advanced Detectors, Nuclear Inst. and Methods in Physics Research, A.

Conference Organization

- Co-organizer of the USA ALICE meeting in Apr 2022 at KU, Lawrence, Kansas
- Co-organizer of the UPC conference in December 2023 in Playa del Carmen, Mexico.

Training Schools

- 2019 CERN-Fermilab HCP Summer School, Geneve (CH).
- Workshop on Nuclear instrumentation for gamma detection – Advancements in Nuclear Instrumentation Measurement Methods and their Applications (ANIMMA), Portoroz (SL).
- Hadron Collider Physics Summer school - HASCO 2015, Gottingen (GE).

PUBLICATIONS

Few-authored publications to which I provided substantial contributions

- **Application of the VMM ASIC for SiPM-based calorimetry**, I. Bearden, V. Buchakchiev, and T. Isidori, et al., arXiv:2403.14577, submitted for publication to JINST
- **Performance of the electromagnetic and hadronic prototype segments of the ALICE Forward Calorimeter**, M. Aehle and J. Alme and T. Isidori, et al., DOI 10.1088/1748-0221/19/07/P07006, JINST
- **Technical Design Report of the ALICE Forward Calorimeter (FoCal)**, ALICE Collaboration, Main contribution to the writing and editing of the document, `https://cds.cern.ch/record/2890281`
- **Performance of a low gain avalanche detector in a medical linac and characterization of the beam profile**, T. Isidori, P. McCavana, B. McClean, R. McNulty, N. Minafra, N. Raab, L. Rock, C. Royon, DOI 10.1088/1361-6560/ac0587, Physics in Medicine and Biology
- **A novel technique for real-time ion identification and energy measurement for in situ space instrumentation**, F. Gautier and A. Greeley and S. G. Kanekal and T. Isidori and G. Legras and N. Minafra and A. Novikov and C. Royon and Q. Schiller, `https://www.sciencedirect.com/science/article/abs/pii/S0168900221005842?via%3Dihub`, Nuclear Inst. and Methods in Physics Research, A
- **Test beam characterization of sensor prototypes for the CMS Barrel MIP Timing Detector**, C. Pena, T. Isidori on behalf of the CMS Collaboration, DOI:10.1088/1742-6596/1162/1/012035, 2021 JINST 16 P07023
- **Diamond detectors for proton Time of Flight in CT-PPS and TOTEM experiments**, T. Isidori, coSupervisor: N.Turini, Supervisor: Simone Donati, Master Thesis: Università degli studi di Pisa, CERN-THESIS-2017-487
- **Timing Performance of a Double Layer Diamond Detector**, M. Berretti, E. Bossini, M. Bozzo, V. Georgiev, T. Isidori, R. Linhart, N. Turini, DOI: 10.1088/1748-0221/12/03/P03026 , JINST 12 (2017) 03, P03026
- **Test of UFSD Silicon Detectors for the TOTEM Upgrade Project**, R. Arcidiacono, M. Berretti, E. Bossini, M. Bozzo, N. Cartiglia, M. Ferrero, V. Georgiev, T. Isidori, R. Linhart, N. Minafra, M. M. Obertino, V. Sola, N. Turini, DOI:10.1088/1748-0221/12/03/P03024, JINST 12 (2017) P03024

Author of ALICE collaboration papers (co-author of more than 100 collaboration papers since 2022)

Author of CMS collaboration papers (co-author of more than 300 collaboration papers since 2017)

CONFERENCES CONTRIBUTIONS, SEMINARS AND POSTERS

ALICE Forward Calorimeter - physics program and expected performance, Quark Matter 2025 in Frankfurt, GE. *2025*

Electron Compton polarimetry R&D, Joint 2024 Electron-Ion Collider User Group (EICUG) meeting in Betlehem, PA. *2024*

FoCal Test Beam results, KU Nuclear Physics Seminar, KU Nuclear Physics Seminar in Lawrence, KS. *2023*

FoCal May Test Beam results, ALICE Week *2023*

FoCal test beam Analysis summary, 3rd ALICE Upgrade Week at CERN *2023*

The FoCal detector at the ALICE collaboration, 2nd International Workshop on Forward Physics and Forward Calorimeter Upgrade in ALICE in Tsukuba, Japan *2023*

FoCal: a high-granularity forward calorimeter at the ALICE experiment, CPAD Workshop. Hosted by Stony Brook University *2022*

Fast detectors for polarimetry at the EIC, American Physics Society in New York (NY), USA *2022*

Fast detectors for polarimetry at the EIC, Forward QCD: open questions and future directions. The university of Kansas graduate seminar in Lawrence, Kansas *2022*

Low Gain Avalanche Detectors for medical applications, Particle Physics on the plain series. Talk recorded and can be found at this link. Hosted by The University of Kansas *2021*

Timing detectors at the EIC. New generation 4D reconstruction and polarimetry, Forward Physics and QCD with LHC, EIC and Cosmic Rays. Hosted by JLab *2021*

Performance of an Ultra Fast Silicon Detector in high rate facilities, UCD - Forward physics meeting in Dublin, Ireland *2019*

Performance of an Ultra-Fast Silicon Detector used to monitor a high rate medical LINAC. Advancements in Nuclear Instrumentation Measurement Methods and their Applications (ANIMMA) in Portoroz, Slovenia. Poster and presentation *2019*

Overview on Timing detectors. Particle physics and other applications at the Kansas Academy of science in Overland Park, Kansas *2019*

Tests of an LGAD detector using a Linac for radiotherapy, Torino PicoSecond Workshop. Torino, Italy *2018*

Radiation effect on Ultra Fast Silicon Detector. TOTEM Collaboration Meeting in Isola d'Elba, Italy *2018*

Timing Performance of a Double Layer Diamond Detector, 5th Beam Telescopes and Test Beams Workshop. Barcelona, Spain *2017*

MENTORSHIP AND TEACHING EXPERIENCE

Organized and lead the training of expert On-calls and shifters for the ALICE EMCAL detector. *2024–present*

Mentor and guide Ph.D., Master, and Bachelor students from the ALICE KU group, maintaining daily and weekly remote communication. *2021–present*

Mentored and guided the work of summer students and interns selected through the CERN Summer Internship Program. *2022–2023*

Mentored fellow graduate students, providing guidance on lab tools and standard procedures for working with solid-state detectors. *2017–2022*

Served as a Teaching Assistant for two laboratory classes for undergraduate honors students at KU. *2022*

Mentored Bachelor students during my Master's internship program at CERN. *2017*

PERSONAL SKILLS AND COMPETENCES

Technical skills

- Expertise in high-energy particle detector technologies and their applications.
- Proficient in data analysis software, including ROOT, C++, Python, and MATLAB.
- Extensive experience in particle beam tests and detector calibration.
- Skilled in detector design, prototyping, and laboratory testing.

Soft skills

- Teamwork: I collaborate daily with multiple teams within the ALICE experiment. I coordinate large groups of collaborators during test beam campaigns, providing support and serving as the local contact at CERN. Additionally, I mentor small groups of students on a weekly and daily basis, guiding them through ongoing projects.
- Working under pressure: Throughout my physics career, I have been deeply involved in testing instrumentation for upgrade projects. This has required me to manage and resolve multiple issues simultaneously while maintaining clear communication and strong collaboration with the teams involved.
- Communication: I co-chair meetings focused on the performance measurement of the ALICE FoCal prototype, offering feedback to teams involved in data analysis and guiding organizational procedures. Additionally, I facilitate communication between the ALICE operation teams and the FoCal design groups.
- Music Studies: I have studied guitar for over 15 years and have experience performing in front of audiences.
- Martial Arts: I have been practicing Judo for over 20 years, achieving the rank of instructor (3rd degree black belt). Additionally, I study Brazilian Jiu-Jitsu and have attained the rank of brown belt. During my time in the USA, I taught both arts to kids and adults, and I am currently the grappling instructor for the CERN Martial Arts Club.

Languages

- Italian: Native speaker.
- English: Very advanced proficiency. I lived in the USA for over 4 years, where I also taught physics classes to undergraduate students.
- Spanish: Basic proficiency with good comprehension.
- French: Basic proficiency, currently studying the language.